

Systematic Options: The Wheel Strategy on NSE Derivatives

What this report covers. Part A tests the wheel on 11 NSE stock options. Part B tests a wheel-like strategy on NIFTY options and futures. Both use public daily market data through 2026-06-30. The stock results account for recorded dividends and corporate actions. The data and modelling limits are explained below.

1. Universe selection

I ranked NSE stocks with listed options by their average daily option turnover over the first 20 trading days of 2020. The top ten form the main universe. This ranking ends on 28 January, before the backtest starts on 29 January, so later trading activity does not affect the ranking.

Rank	Ticker	Avg option turnover	Avg OI contracts
1	RELIANCE	INR 4,862.70 crore/day	38,924
2	SBIN	INR 4,466.73 crore/day	28,724
3	INDUSINDBK	INR 2,566.69 crore/day	22,591
4	ICICIBANK	INR 2,126.44 crore/day	18,643
5	INFY	INR 2,064.53 crore/day	14,005
6	AXISBANK	INR 2,026.78 crore/day	12,058
7	BHARTIARTL	INR 1,618.37 crore/day	8,620
8	BAJFINANCE	INR 1,606.77 crore/day	9,539
9	TCS	INR 1,580.28 crore/day	20,384
10	KOTAKBANK	INR 1,574.06 crore/day	12,837

ADANIENT is the eleventh stock, added by hand to test a more volatile name. Because the assignment asks for rule-based selection, the full 11-stock list meets that requirement only partly. Its later price path was already known when it was added, which can bias the result. Treat this as a stress-test example, not an unbiased test of stock selection.

The top-15 ranking is in data/universe_top15_liquidity.csv. The first-day trading-activity check for the 11 stocks is in data/universe_selection.csv.

2. Results at a glance

All three portfolios start with INR 20 million. NIFTY 50 total return includes reinvested dividends. The 11-stock buy-and-hold benchmark includes recorded dividends and corporate-action benefits.

Series	End value	CAGR	Annual vol	Sharpe	Sortino	MDD	Calmar
Wheel portfolio	INR 36,690,281	9.92%	12.47%	0.32	0.28	-29.00%	0.34
NIFTY 50 total return	INR 42,460,973	12.45%	18.05%	0.39	0.36	-37.46%	0.33
Universe buy-and-hold	INR 62,335,789	19.38%	26.36%	0.57	0.52	-43.98%	0.44

CAGR is average yearly growth. Annual volatility measures how much daily returns vary. Sharpe and Sortino compare return with risk; higher is better. MDD is the largest fall from a previous peak. Calmar compares yearly growth with that largest fall.

Series	Peak / MDD start	Trough	Recovery
Wheel portfolio	2020-02-20	2020-03-23	2020-11-09
NIFTY 50 total return	2020-02-12	2020-03-23	2020-11-05
Universe buy-and-hold	2020-02-20	2020-03-23	2020-11-09

Main result. The stock wheel grew at 9.92% a year, compared with 12.45% for NIFTY 50 total return and 19.38% for buying and holding the same 11 stocks. Its day-to-day swings and worst loss from a peak were smaller, but its Sharpe ratio was below NIFTY 50. Option premiums did not make up for the return it missed. The March 2020 crash shows how short puts can lose sharply when stocks fall.

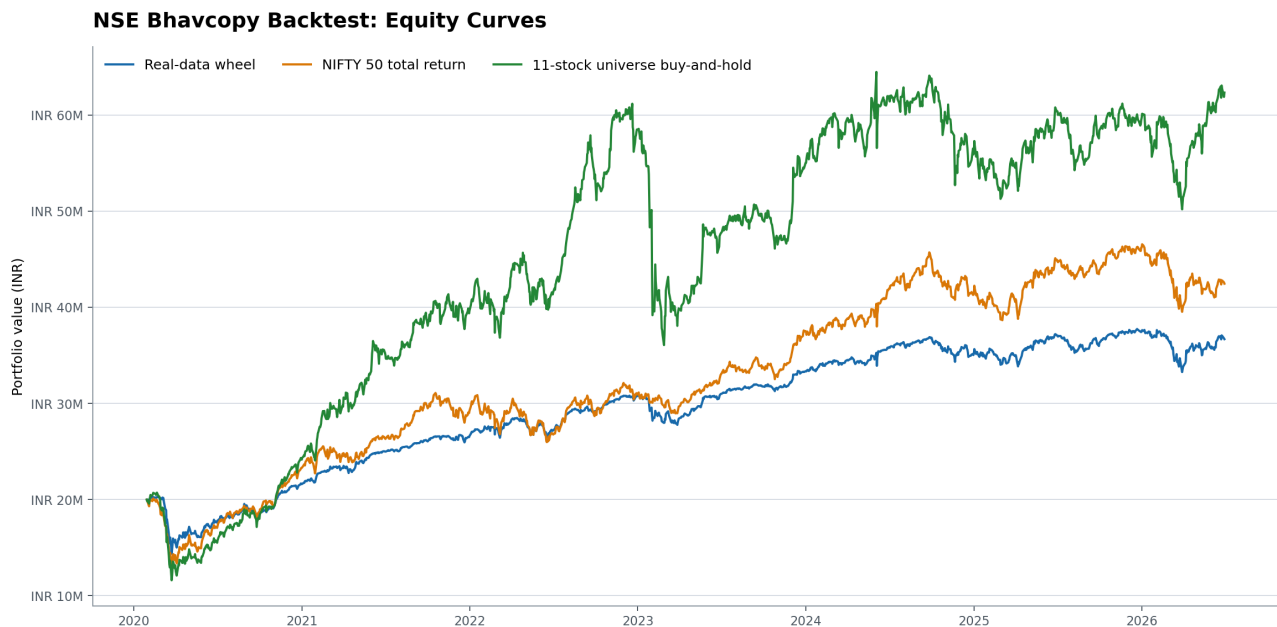


Figure 1. Portfolio equity curves from the real-data run.

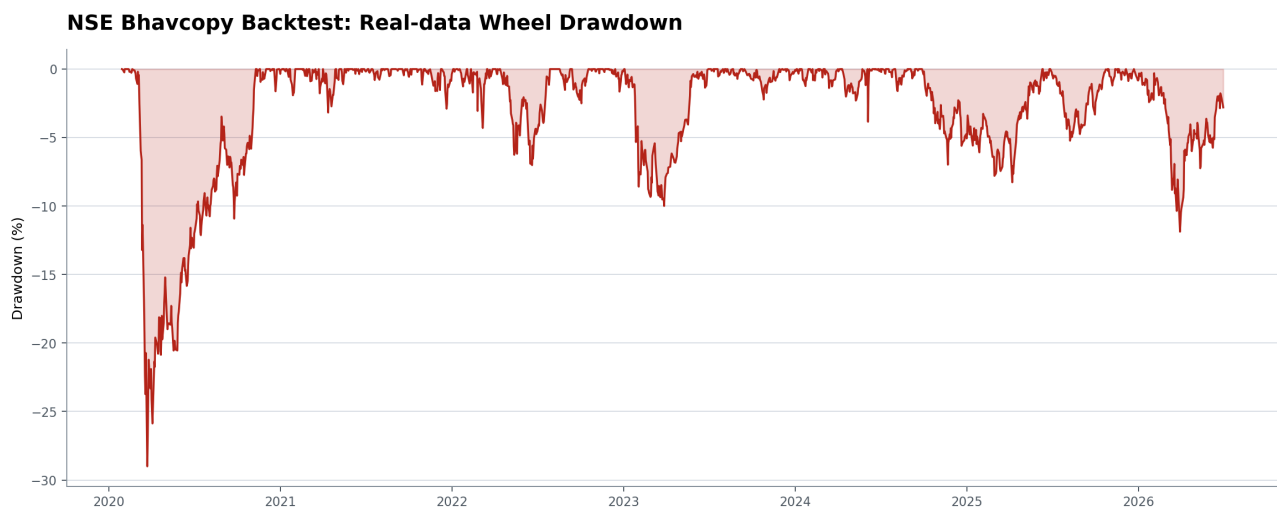


Figure 2. Drawdown calculated from the wheel equity curve.

3. Part A: how the stock wheel works

Capital and entry. INR 20 million is divided equally among 11 stocks. For each new trade, the strategy uses yesterday's stock close to choose a strike and today's recorded option close as the assumed sale price. It requires a positive option price, trading volume, and open interest. It chooses the nearest listed expiry that is at least seven calendar days away, and trades only whole exchange lots. Unused cash earns no interest in the base case. The 6.5% rate in Sharpe and Sortino is used only to judge performance; it is not added to the account.

The cycle. When a stock sleeve holds cash, it sells a put with a strike near 5% below the reference stock price and keeps enough cash to buy all shares if assigned. If the stock ends below the strike at expiry, the sleeve buys those shares at the strike. It then sells covered calls near 5% above the reference price, but never below its net purchase cost. If the stock ends above the call strike, the shares are delivered and the sleeve returns to cash. Otherwise it keeps the shares and tries another call. The strategy holds options to expiry; it has no early exit, stop, or roll rule. A mandatory delivery pays delivery costs. The margin estimate rises from 20% to 35% of the option's value when delivery risk is high in the final seven days.

Why these rules? A 5% strike rule can be repeated without estimating option delta or implied volatility; the 3% and 8% tests show how much results depend on that choice. Monthly expiries avoid frequent trading. Holding to expiry avoids assuming an unknown intraday exit price. The call price floor avoids knowingly selling shares below their net purchase cost, although it can leave a losing stock without an eligible call. Cash security and whole-lot rounding keep positions within the INR 2 crore account.

The trade log has **1,503 events**: 359 put sales, 89 put assignments, 391 call sales, and 79 call assignments, along with other events. Lot size had to be inferred for 552 entries. A missing current option price led to 22 carried-forward marks, and 10 listed expiries changed.

Part A decision rules

Decision	Rule used by the engine
Reference price	Yesterday's stock close, adjusted if a corporate action takes effect today
Expiry	Nearest listed expiry with at least 7 calendar days left
Put strike	Listed strike closest to 95% of the reference price, and below it
Call strike	Listed strike closest to 105% of the reference price, but not below net share cost
Option data	Recorded close, volume, and open interest must all be above zero
Sale / daily value	Assume a sale at today's option close; use NSE settlement to value it later
Position size	Round cash-backed puts and share-backed calls down to whole exchange lots
At expiry	A put below its strike buys shares; a call above its strike delivers shares

Example: RELIANCE closed at INR 1,479.85 on 29 January 2020. Five percent below that was INR 1,405.86. On 30 January, the strategy sold the available INR 1,400 put for INR 24.95 per share. Its INR 1.818 million allocation covered two lots of 500 shares. On 27 February the put finished in the money, so it bought 1,000 shares at INR 1,400. After the premium, the net share cost was INR 1,375.05 each, before delivery fees.

Ticker	CSP entries	Assign %	Call %	Avg days	Capture	Premium	Costs
RELIANCE	36	22.22%	16.67%	158.6	13.42%	INR 1,451,106	INR 43,452
SBIN	41	24.39%	27.27%	112.1	-69.35%	INR 2,303,703	INR 61,067
INDUSINDBK	9	22.22%	2.50%	1183.0	68.46%	INR 562,495	INR 9,220
ICICIBANK	47	12.77%	17.24%	124.8	20.77%	INR 1,378,905	INR 31,983
INFY	24	16.67%	6.98%	336.0	-30.88%	INR 1,072,890	INR 21,661
AXISBANK	23	39.13%	16.33%	195.6	-63.14%	INR 1,726,034	INR 58,404
BHARTIARTL	44	20.45%	25.00%	103.0	37.62%	INR 1,891,273	INR 54,668
BAJFINANCE	48	27.08%	54.17%	67.9	-46.97%	INR 2,257,475	INR 80,428
TCS	31	32.26%	28.12%	84.9	11.07%	INR 1,446,412	INR 60,864
KOTAKBANK	32	40.62%	26.67%	105.4	-62.77%	INR 1,413,480	INR 66,541
ADANIENT	24	20.83%	18.18%	92.8	-13.88%	INR 2,525,070	INR 31,853

CSP entries counts every put sale, including contracts still open at the end. Avg days measures how long shares were held in cycles that ended with a call assignment. Capture compares option profit at expiry with the premiums received on completed contracts. It can be negative when assignment losses exceed premiums.

Risk and return by stock

Ticker	CAGR	Vol	Sharpe	Sortino	MDD	Peak	Trough	Recovery	Calmar
RELIANCE	8.78%	15.83%	0.21	0.20	-27.50%	2020-02-20	2020-03-23	2020-04-30	0.32
SBIN	13.13%	19.70%	0.41	0.34	-43.69%	2020-02-27	2020-05-22	2021-01-07	0.30
INDUSINDBK	-0.00%	26.31%	-0.11	-0.11	-56.19%	2020-02-06	2020-03-25	2021-09-16	-0.00
ICICIBANK	9.81%	16.00%	0.27	0.23	-29.29%	2020-03-03	2020-03-23	2020-11-13	0.33
INFY	2.33%	18.41%	-0.13	-0.12	-31.79%	2025-01-10	2026-06-30	not yet	0.07
AXISBANK	14.06%	24.30%	0.41	0.38	-48.04%	2020-03-03	2020-03-24	2021-01-08	0.29
BHARTIARTL	13.09%	13.35%	0.52	0.48	-21.35%	2020-09-02	2020-10-19	2020-12-28	0.61
BAJFINANCE	14.43%	16.61%	0.52	0.46	-33.40%	2020-03-03	2020-05-26	2020-11-20	0.43
TCS	4.94%	14.71%	-0.03	-0.03	-42.88%	2024-12-13	2026-06-30	not yet	0.12
KOTAKBANK	10.14%	15.59%	0.29	0.27	-20.71%	2020-03-05	2020-03-23	2020-10-27	0.49
ADANIENT	12.66%	34.02%	0.34	0.30	-59.27%	2023-01-09	2023-02-27	2024-06-03	0.21

Each stock starts with INR 20 million divided by 11. 'Not yet' means its earlier peak had not been regained by the final date. Exact values are in outputs_real/per_name_summary.csv.

Where the profit and loss came from

P&L component	Portfolio INR
Option premiums	INR 18,028,843
Stock delivery + mark	INR -1,661,978
Dividends	INR 769,820
Rights value	INR 73,736
Spun-off stock	INR 0
Cash interest	INR 0
All costs	INR -520,141
Open option liability	INR 0
Ending equity (incl. INR 20m start)	INR 36,690,281

Profit on completed share deliveries was INR 3,605,810. It is already included in the stock line, which also values shares still held at the end; adding it again would count it twice.

Profit and loss by stock

Ticker	Premium	Stock P&L	Dividends + actions	Costs	Ending equity
RELIANCE	INR 1,451,106	INR -141,878	INR 36,000	INR -43,452	INR 3,119,958
SBIN	INR 2,303,703	INR -181,800	INR 134,850	INR -61,067	INR 4,013,867
INDUSINDBK	INR 562,495	INR -587,780	INR 34,200	INR -9,220	INR 1,817,877
ICICIBANK	INR 1,378,905	INR 148,920	INR 0	INR -31,983	INR 3,314,024
INFY	INR 1,072,890	INR -983,520	INR 222,000	INR -21,661	INR 2,107,891
AXISBANK	INR 1,726,034	INR 738,750	INR 5,525	INR -58,404	INR 4,230,086
BHARTIARTL	INR 1,891,273	INR 341,143	INR 8,550	INR -54,668	INR 4,004,480
BAJFINANCE	INR 2,257,475	INR 281,250	INR 41,000	INR -80,428	INR 4,317,479
TCS	INR 1,446,412	INR -1,007,062	INR 280,875	INR -60,864	INR 2,477,542
KOTAKBANK	INR 1,413,480	INR 214,000	INR 720	INR -66,541	INR 3,379,841
ADANIENT	INR 2,525,070	INR -484,000	INR 79,836	INR -31,853	INR 3,907,234

Costs appear as negative amounts. The full calculation, including any open option value, is in `outputs_real/pnl_attribution.csv`.

4. Data and costs

The data collection checked weekdays from 2020-01-01 through 2026-06-30. It found 1,604 stock-price dates, 1,603 option dates, and 4,380,500 option rows. On 91 weekdays none of the three NSE reports was available; one date had prices but no option report.

Data sources

Data source	What it provides	Main limitation
NSE stock and index reports	Daily closes for 11 stocks and NIFTY 50	No intraday prices or guaranteed trade price
NSE option reports	Daily option close, settlement, volume, open interest and lot size	No bid/ask prices; many contracts did not trade
Corporate-action notices	Dates and amounts for dividends, splits, bonuses, rights and demergers	Extra public records were needed for two stocks
NIFTY 50 total-return index	Official index value with dividends reinvested	An investor would need a tracking fund to follow it

Coverage check	Value
Price dates	1,604
Option dates	1,603
Option rows	4,380,500
Duplicate contract keys	0
Price cells missing	0
Missing bid/ask fields	4,380,500 / 4,380,500
Zero volume fields	2,848,434

The strategy chooses strikes using yesterday's stock close and assumes options sell at today's recorded close. Lot sizes come from NSE data or a checked estimate for older contracts. Dividends and corporate actions are applied on their effective dates.

Puts are backed by cash and calls by held shares. The daily margin check uses an estimate of 20% of option value, rising to 35% near expiry when delivery is likely; held shares count at 80% of market value. These rates are assumptions, not actual exchange margin records. Full settings are in `outputs_real/assumptions.json`.

Transaction-cost model

Cost item	Applied rule
Brokerage	INR 20 for every executed option, futures or delivery leg
Option exchange / tax	0.035% of premium exchange charge; 18% GST on brokerage + exchange; SEBI INR 10/crore
Option-sale STT	0.050% before Apr-2023; 0.0625% to Sep-2024; 0.100% to Mar-2026; 0.150% thereafter
Option slippage	25 bps of premium in the base case; 10 and 50 bps sensitivity cases
Stamp duty	Assigned share purchases pay 0.015% of delivery value (constant-rate approximation); futures buys pay 0.002%; option entries are sells
Physical stock delivery	0.1% STT plus delivery charges; mandatory strike delivery has no slippage
Market stock sales	Delivery costs plus 25 bps of notional for residual and spun-off-share sales
Futures legs	0.00173% exchange charge, GST, SEBI fee, 2 bps notional slippage; buy stamp duty 0.002%
Futures-sale STT	0.010% before Jun-2023; 0.0125% to Sep-2024; 0.020% thereafter
Futures roll	Charged as two executions: sell expiring contract and buy next contract

Total estimated trading costs are INR 520,141 for the stock wheel and INR 494,197 for the NIFTY wheel. STT is India's securities transaction tax. GST is the goods and services tax. A basis point (bp) is 0.01%. The buyer of an exercised option pays option-exercise STT; this short-option strategy pays share-delivery STT when shares move.

A split or bonus changes the number of shares and option units, while prices and cost per share adjust in the opposite direction. For a rights issue, the strategy uses the NSE strike and lot changes and adds the estimated value of the rights. A demerger creates shares in the new company; they are valued at an estimate until listing and then sold by the wheel at the first recorded close. A dividend is paid only when shares were held before the ex-date. Contract-expiry and symbol changes are logged.

Corporate-action screen for all 11 names

Ticker	Recorded event(s)	Largest adjusted 1-day drop
RELIANCE	2020-05-13 RIGHTS; 2023-07-19 EXPIRY_RESET; 2023-07-20 SPIN_OFF; 2024-10-28 BONUS	13.15% (2020-03-23)
SBIN	No large action found	14.4% (2024-06-04)
INDUSINDBK	No large action found	27.16% (2025-03-11)
ICICIBANK	No large action found	17.85% (2020-03-23)
INFY	No large action found	10.04% (2020-03-23)
AXISBANK	No large action found	27.91% (2020-03-23)
BHARTIARTL	2021-09-27 RIGHTS	11.95% (2020-03-23)
BAJFINANCE	2025-06-16 SPLIT_BONUS	23.23% (2020-03-23)
TCS	No large action found	9.41% (2020-03-12)
KOTAKBANK	2026-01-14 SPLIT	13.0% (2020-03-23)
ADANIANT	2025-11-17 RIGHTS	28.2% (2023-02-01)

This check compares the stock price before and after each day, accounting for recorded corporate actions. It can flag a large unexplained drop, but it cannot prove that every small payment or adjustment was captured. Dividends come from saved NSE records and documented public additions.

Data limits. The option files have no bid or ask prices, so a recorded close may not have been available for the strategy to trade. The 93 dividend records in `data/dividends.csv` are credited only when shares were already held. Rights use NSE estimates. Shares from a demerger use an estimated value until they list. Actual daily exchange margin files were not used; the estimated margin and account balances are in `outputs_real/daily_positions.csv`.

5. Sensitivity and stress narrative

Scenario	Strike OTM	Slippage	CAGR	Vol	Sharpe	MDD
Base: 5% target	5.00%	25 bps	9.92%	12.47%	0.32	-29.00%
3% target	3.00%	25 bps	9.67%	12.77%	0.29	-28.17%
8% target	8.00%	25 bps	9.12%	11.83%	0.26	-28.07%
Low trading friction	5.00%	10 bps	9.93%	12.46%	0.32	-28.99%
High trading friction	5.00%	50 bps	9.89%	12.48%	0.31	-29.01%

Changing the target strike from 5% to 3% below or above the stock changes the Sharpe ratio from 0.32 to 0.29; using 8% changes it to 0.26. Changing assumed slippage from 10 to 50 basis points (0.10% to 0.50%) changes annual return only slightly. That does not prove real trades would fill at these prices because the data has no bid or ask quotes.

The wheel's largest fall from an earlier peak was -29.00%, from 2020-02-20 to 2020-03-23. It regained that peak on 2020-11-09. Short puts were already open when the fall began. As stocks dropped, puts on INDUSINDBK, RELIANCE ended in the money and forced share purchases. The value of shares held rose from INR 0 to INR 1,287,790. The five largest losses by stock are below.

Sleeve	Peak position	Trough position	Stock move	Equity change
INDUSINDBK	0 shares; short put 1200	1200 shares; short call 1200	-71.54%	INR -967,502
AXISBANK	0 shares; short put 700	0 shares; short put 700	-58.53%	INR -865,349
ADANIENT	0 shares; short put 220	0 shares; short put 220	-53.26%	INR -684,781
SBIN	0 shares; short put 300	0 shares; short put 310	-44.58%	INR -652,641
ICICIBANK	0 shares; short put 500	0 shares; short put 490	-48.08%	INR -523,218

INDUSINDBK had the largest change in account value, INR -967,502, while its stock moved -71.54%. Stocks bought through assignment then kept falling. During this period the strategy made 11 new option sales and paid INR 5,850 in recorded event costs. Daily cash, shares, and open option values are in outputs_real/daily_positions.csv.

6. Part B: implemented synthetic NIFTY wheel

NIFTY options pay cash at expiry; they cannot deliver the index itself. To imitate the stock wheel, the strategy sells a monthly put backed by cash. If it ends in the money, the option pays its loss in cash and the strategy buys NIFTY futures on the next trading day. It then sells calls against the same index exposure. If a call ends in the money, it pays cash and the futures are sold. Otherwise the futures stay open and move to a new contract at expiry. The call strike cannot be below the price needed to recover the earlier put loss. Futures gains and losses are recorded each day, and each trade pays estimated costs.

Series	End value	CAGR	Vol	Sharpe	Sortino	MDD	Calmar
Synthetic NIFTY wheel	INR 27,527,240	5.04%	12.47%	-0.06	-0.05	-24.74%	0.20
NIFTY 50 total return	INR 42,283,722	12.22%	17.97%	0.38	0.35	-38.27%	0.32

The synthetic wheel's worst fall ran from 2020-03-05 to 2020-04-03 and recovered on 2020-07-09. NIFTY total return fell most from 2020-01-14 to 2020-03-23 and recovered on 2020-11-06.

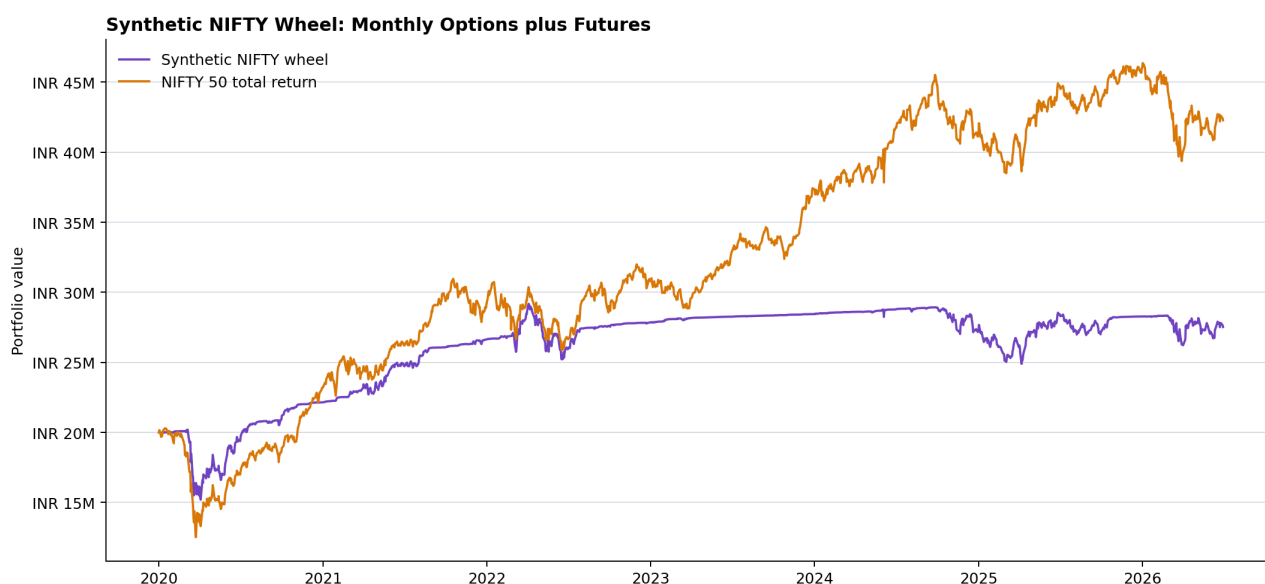


Figure 3. Implemented monthly synthetic NIFTY wheel versus NIFTY 50 total return.

Part B decision logic

Decision	Rule used by the engine
Contracts	Monthly NIFTY options that end when listed futures end; weekly options excluded
Strike	Use yesterday's NIFTY close: put near 95%; call at least 105% and above recovery cost
Trade / daily value	Assume today's recorded close for entry; use settlement prices for later option values
After put loss	Pay cash on the put and buy futures next session for the same index units
Recovery cost	Futures purchase price + put loss per unit - original put premium per unit
Futures gains/losses	Add or subtract each day's futures price change directly from cash
After call expiry	An in-the-money call pays cash and futures are sold; otherwise futures continue or roll
Account value	Cash less the value of the short option; futures gains/losses are already in cash

From 1 January 2020 to 30 June 2026, the strategy sold 48 puts; 12.50% ended in the money and led to futures purchases. It sold 30 calls; 16.67% ended with a futures sale. Futures moved to a new contract 24 times. On 15 option dates and 1 futures dates, a missing current value was replaced by the last available settlement. The strategy reserves enough cash to cover every put. While it holds futures and a short call, its estimated margin is the larger of 15% of futures value and 20% of call exposure. Security cash earns no interest in this base case.

Part A versus Part B mechanics and risk

Issue	Stock wheel	Synthetic NIFTY wheel
Assignment	Physical shares move at strike	Option pays cash intrinsic; futures bought next session
Main risk	A stock can fall suddenly; company events also matter	One company matters less, but a market crash affects all exposure
Timing	The option and delivered shares use the same stock	NIFTY can move before futures are bought next day; futures can differ from the index
Roll	No stock roll after delivery	24 futures rolls; each pays two legs; cumulative raw calendar spread 1,804.15 index points
Margin	Cash backs puts; held shares back calls at 80% of value	Uses the larger of 15% futures or 20% call value; actual margin may differ
Lot size	Each stock allocation rounds down to its own exchange lot	NIFTY units round down when the exchange lot changes
Expiry choice	Nearest monthly option with at least 7 days left	Monthly options match futures; weekly trading would require more trades

A futures price can differ from the NIFTY index. The strategy can lose if NIFTY moves between option expiry and the next day's futures purchase, or when it switches from an expiring future to a more expensive new one. The estimated margin credit for holding related positions lowers required cash, but it does not remove market risk. Using an index reduces single-company risk while retaining the risk of a broad market fall.

Part B ledger and sensitivity

Reconciled component	INR
Starting capital	INR 20,000,000
Option premium	INR 6,866,404
Option cash settlement	INR -8,840,078
Futures variation margin	INR 9,995,112
Cash interest	INR 0
Costs	INR -494,197
Ending equity	INR 27,527,240

Scenario	OTM	Opt slip	Fut slip	CAGR	Sharpe	MDD
Base	5.00%	25 bps	2 bps	5.04%	-0.06	-24.74%
3% target	3.00%	25 bps	2 bps	6.59%	0.07	-29.04%
8% target	8.00%	25 bps	2 bps	4.77%	-0.10	-27.23%
Low option friction	5.00%	10 bps	2 bps	5.05%	-0.06	-24.74%
High option friction	5.00%	50 bps	2 bps	5.03%	-0.06	-24.74%
High futures friction	5.00%	25 bps	4 bps	4.89%	-0.07	-24.75%

The base NIFTY strategy earned less than buying and holding NIFTY total return, but its returns moved less and its worst fall was smaller. Choosing a 3% strike target improved return but increased the worst fall; an 8% target brought in less premium and earned less. Futures slippage has a greater effect than option slippage because it is charged on the full futures value.

At option expiry, the model calculates the cash payment from that day's NIFTY close because the archive does not provide a consistent separate final-settlement index value. It buys futures at the next day's recorded close, not the open. Actual bid/ask prices and past daily exchange margin files are unavailable. Only monthly options were used; weekly trading would create more trades and more risk close to expiry.

7. Assignment checklist and limits

Assignment item	Report evidence	Status
2020-01-01 to 2026-06-30 EOD data	Section 4; 20-session formation window; backtest starts 2020-01-29	Covered
Objective 8-15 stock liquidity screen	Section 1; top 10 by turnover; ADANIENT is a disclosed override	Partial
Put, delivery, call, sizing, exits and margin	Section 3 rules, decision table and worked RELIANCE example	Covered
Costs, STT and slippage sensitivity	Section 4 cost table; Section 5 sensitivity	Covered
Portfolio and per-name metrics / diagnostics	Sections 2-3 tables and reconciled P&L	Covered
Benchmarks and worst-drawdown narrative	NIFTY 50 TR, universe buy-and-hold, and March 2020 walk-through	Covered
Synthetic-index design and comparison	Section 6 mechanics, basis, roll, margin, lots and risk comparison	Covered
Biases, sources, verdict and reproducibility	Section 7 and machine-readable output files	Covered

Daily stock and option prices come from NSE reports. Dates with missing files are listed in data/source_manifest.json. Dividend and corporate-action sources appear beside each event in data/dividends.csv and data/corporate_actions.csv. NIFTY total return comes from NSE Indices. Tax rules are based on the NSE tax guide, and margin context comes from the NSE margin FAQ.

What could make the results look too good? The top-ten turnover ranking uses only days before trading starts. ADANIENT was added after its later behaviour was known, however, and choosing stocks with data through June 2026 can leave out companies that disappeared. The model chooses from today's option contracts and assumes it can trade at today's closing price, even though the file has no bid or ask quotes. Missing option prices carry forward the last settlement. Some corporate actions or dividends may be missing. Margin uses fixed estimates rather than actual past exchange requirements. The stock test starts on 29 January, while the NIFTY test starts on 1 January, so their annual returns do not cover exactly the same dates.

Can this be traded as shown? The stock wheel earned 9.92% a year and the NIFTY futures wheel earned 5.04%, compared with 12.45% for NIFTY 50 total return. Neither strategy earns interest on cash held as security. These results are useful for studying the rules, but executable prices and actual historical margins are needed before claiming the strategy is ready for live trading.

To rebuild the results, install requirements.txt, then follow the commands in README.md. The main scripts generate the data, backtests, charts, and this report in that order. Automated tests are under tests/.